

Yashodhan Zingade

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PROFESSIONAL SUMMARY

Cybersecurity and Machine Learning professional hands-on experience in penetration testing, vulnerability assessment, SOC operations. Proficient in network security, OSINT, incident detection, and secure system design, additional experience in blockchain development, Li-Fi communication systems, and **Industrial Control Systems (ICS) / Operational Technology (OT) security**. Top 3 Percent globally on TryHackMe, consistent record of identifying attack vectors, automating security workflows.

EDUCATION

Veermata Jijabai Technological Institute (VJTI), Mumbai

Jun 2023 – Jun 2027

Bachelor of Technology in Electrical Engineering

Minor: Electric Vehicles, Cybersecurity

CGPA: 8.36/10

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, Solidity, Bash, Rust

Machine Learning: Scikit-learn, TensorFlow, Pandas, Data Analysis, Model Training, Feature Engineering

Cybersecurity: Penetration Testing, Vulnerability Assessment, OSINT, NIST Framework, OWASP Top 10, SOC Operations

Security Tools: Burp Suite, CALDERA, Docker, Malcolm, Nmap, Suricata, Wireshark

Engineering Tools: GitHub, MATLAB, Simulink, VS Code

Cloud & Platforms: Google Colab, Azure Machine Learning, REST APIs

Databases: MySQL, MongoDB, Valkey

EXPERIENCE

COE-CNDS – Intern

Jun 2026 – Aug 2026

- Secured Siemens Smart Grid ICS/OT environments by implementing 10+ threat detection workflows using **Suricata**, **Malcolm**, and **CALDERA**, mapping security events to the MITRE ATTCK for ICS framework.
- Investigated 50+ security alerts and network events, supporting threat detection, incident response, and technical documentation in compliance with NDA requirements.
- Synthesized 30+ research papers through a structured literature survey, building applied expertise in 15+ AI/ML concepts including RAG, LLMs, Llama, CNN-LSTM, Reinforcement Learning, Transfer Learning, and **Explainable AI (XAI)** for OT/ICS environments.

Cybersecurity Analyst

Aug 2025 – Sept 2025

- Conducted vulnerability analysis aligned with the OWASP Top 10, identifying attack surfaces and improving detection strategies within SOC workflows.
- Penetration testing assessments on 10+ web applications, identifying, reporting critical vulnerabilities with recommendations.
- Analyzed 2K+ daily security events to detect anomalies, reducing false positives through effective alert triage and investigation.

PROJECTS

Smart India Hackathon Platform — Blockchain, Machine Learning, Full Stack

Dec 2024

- Engineered a secure digital transaction platform with a scalable backend architecture using Ethereum and blockchain, integrating frontend, backend, and distributed systems for secure data flow.
- Optimized system performance to achieve 110 transactions per second (TPS) with an average transaction latency of 12 seconds.

Li-Fi Research Paper (Documentation) — Visible Light Communication, IoT

Mar 2024

- Designed an LED-based Li-Fi communication system using photodiode receivers, enabling visible light communication with throughput exceeding 2 Gbps and peak performance of 45 Gbps.
- Benchmarked Li-Fi against Wi-Fi across security, latency, and IoT application criteria.

Cybersecurity Toolkit (Malware Simulation) — C, C++, OSINT, Networking

Dec 2024

- Built 20+ offensive security toolkits spanning AES encryption, phishing simulation, enumeration, and additional offensive security sub-domains.
- Implemented offensive security features for network analysis, traffic monitoring, and detection bypass techniques.

Healthcare Blockchain Platform — Ethereum, Solidity, IPFS, FastAPI

Jan 2024

- Built a decentralized healthcare system using smart contracts with secure, role-based access control.
- Achieved 2nd place among 90+ teams at Inheritance 2024.

CERTIFICATIONS

Blockchain Architecture Design and Use Cases – LearnQuest

Blockchain and Cryptocurrency – University of Michigan

Introduction to Cyber Attacks – New York University

Certified Ethical Hacking – Cisco

LEADERSHIP AND ACTIVITIES

HackersEra, Pune

2023 – 2025

- Served as Scrum Master within a cybersecurity community, coordinating project-based freelance cybersecurity associates.